EQ. 9 **Fully expanded formula** · the two levels in one expression

$$\mathrm{MP}_{\mathrm{norm}} = \{[(\Sigma_i\, w_i\, \mathrm{MI}_i) \times (F \times P \times \mathrm{MDA}^{\mu})^{1/(2+\mu)}] \times (\mathrm{PW}^{w_1}\, \mathrm{IW}^{w_2}\, \mathrm{SocW}^{w_3})^{1/\Sigma w} \times [(\mathrm{IC}^{a_1}\, \mathrm{TE}^{a_2}\, \mathrm{BC}^{a_3})^{1/\Sigma a}]^{\gamma}$$
$$\times\, [\min\{1,\, (\beta_1\, \mathrm{CS}_{\mathrm{m}} + \beta_2\, \mathrm{CD}_{\mathrm{m}} + \beta_3\, \mathrm{RC}_{\mathrm{m}})(1 + \delta \cdot \mathrm{CDip})\}]^{\alpha}\}^{1/(3+\gamma+\alpha)}$$

$X_k \to 0$ Collapse

If any component tends to zero, so does $\mathrm{MP}_{\mathrm{norm}}$ (every exponent is positive).

$\partial \mathrm{MP}_{\mathrm{norm}}/\partial X_k$
Increasing and concave

Strictly increasing and strictly concave in each component for any α, γ > 0: decreasing marginal returns everywhere.

$\partial \mathrm{MP}_{\mathrm{norm}}/\partial \alpha$ Sign of CP − G

Monotone in α with the sign of CP − G; as α → 0 the index tends to G and as α → ∞ to CP.

$\mathrm{MP}_{\mathrm{norm}} \in (0, 1]$ Bounded

Between min{G, CP} and max{G, CP}; interpretable through the five-level scale.

Note: substituting Eqs. 4–8 into Eqs. 2–3 yields the complete model. All parameters are calibratable by state without modifying the architecture; the model has been verified numerically (300,000 random draws) but neither calibrated nor externally validated. Verification scripts: verification/verify_model.py and robustness_round2.py.